Date

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Section ⇒ K (CSE)

Course Code ⇒ MAS2001

Assignment ⇒ 3

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Section A

Q A1) The criteria of good estimators are
- unbiasedness, consistency, efficiency, sufficiency

Q A2) Write any two methods of estimation
- Maximum likelihood estimation (MLE)
- Method of moments

Q A3) If MLE exists, it is the most ... in the class of such estimators
- efficient

Q A4) Estimate the parameter p of the binomial distribution by the method of moments

$\mu = np$

variance $\sigma^2 = np(1-p)$

by method of moments, $\hat{p} = \bar{x}/n$

Q A5) Let T_1 and T_2 be stats with expectations $E(T_1) = \theta_1 + \theta_2$ and $E(T_2) = \theta_1 - \theta_2$, find unbiased estimators of θ_1 and θ_2

$E(\theta_1) = \theta_1 + \theta_2$ — ①

$E(\theta_2) = \theta_1 - \theta_2$ — ②

solving ① and ②

$\theta_1 = T_1 + T_2 / 2$

$\theta_2 = T_1 - T_2 / 2$

Q A6) Under what circumstances can the normal distribution be used to find confidence limits of the population mean?

→ when the population is normally distributed

→ for large sample, the CLT ensures the sampling distribution of mean approximately normal

Q A7) The interval ... is known as the 95% confidence interval for μ, and the 95% confidence limits are ...

$\bar{x} \pm 1.96\left(\frac{\sigma}{\sqrt{n}}\right)$, $\bar{x} \pm t_{0.025, n-1}\left(\frac{s}{\sqrt{n}}\right)$

Q A8) Under Bayesian approach, the parameter of a distribution is a ... Random variable

Q A9) Bayes estimator is always a function of ... statistics

→ sufficient statistics

Q A10) Give an example of an estimator

(i) which is consistent but not unbiased

(ii) which is unbiased but not consistent

(i) Consistent but not unbiased:

$$S_n^2 = \frac{1}{n} \sum_{i=1}^{n} (x_i - \bar{x})^2$$

It is consistent but biased for the population variance

(ii) Unbiased but not consistent

$$\bar{u} = x_1$$

as it does not converge to the true mean as the sample size inc.

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Section - B

Q1) Likelihood function for normal distribution

$N(\mu, \sigma^2)$

$$f(x_i; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i-\mu)^2}{2\sigma^2}\right)$$

for sample $x_1, x_2, \dots x_n$

$$L(\mu, \sigma^2; x_1, x_2, \dots, x_n) = \prod_{i=1}^{n} f(x_i; \mu, \sigma^2)$$

$$L(\mu, \sigma^2; x_1, x_2, \dots x_n) = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^{n} (x_i-\mu)^2\right)$$

factorizing

$$\sum_{i=1}^{n} (x_i-\mu)^2 = \sum_{i=1}^{n} (x_i^2 - 2x_i\mu + \mu^2)$$

$$= \sum_{i=1}^{n} x_i^2 - 2\mu \sum_{i=1}^{n} x_i + n\mu^2$$

$$\Rightarrow \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n \exp\left(-\frac{1}{2\sigma^2}\left(\sum_{i=1}^{n} x_i^2 - 2\mu \sum_{i=1}^{n} x_i + n\mu^2\right)\right)$$

$$\Rightarrow \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^{n} x_i^2\right) \exp\left(-\frac{1}{2\sigma^2}\left(-2\mu \sum_{i=1}^{n} x_i + n\mu^2\right)\right)$$

$\therefore \sum_{i=1}^{n} x_i$ is sufficient for μ

$\sum_{i=1}^{n} x_i^2$ is sufficient for σ^2

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Q2) Likelihood function

$$L(\theta; x_1, x_2, \dots x_n) = \prod_{i=1}^{n} 1/\theta = (1/\theta)^n$$

for $0 \le x_i \le \theta$

$$L = (\theta; x_1, x_2 \dots x_n) = (1/\theta)^n \cdot I(x_1, x_2 \dots x_n \le \theta)$$

let $T(x) = \max(x_1, x_2 \dots x_n)$

condition $x_i \le \theta$ is equivalent to $T(x) \le \theta$

$$L(\theta; x_1, x_2 \dots x_n) = (1/\theta)^n \cdot I(T(x) \le \theta)$$

$h(x) = 1$

$g(T(x), \theta) = (1/\theta)^n \cdot I(T(x) \le \theta)$

$\therefore$ sufficient statistic for θ in a uniform distribution on $[0, \theta]$ is the max value in the sample, $T(x) = \max(x_1, x_2 \dots x_n)$

Q3) (i) Likelihood function

$$L(\mu) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(\frac{-(x_i-\mu)^2}{2\sigma^2}\right)$$

log likelihood function

$$\ln L(\mu) = -\frac{n}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^{n} (x_i-\mu)^2$$

$$\frac{d}{d\mu} \ln L(\mu) = \frac{1}{\sigma^2} \sum_{i=1}^{n} (x_i-\mu) = 0$$

$\mu = \frac{1}{n} \sum_{i=1}^{n} x_i$ MLE for μ is sample mean

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(2) likelihood function

$$L(\sigma^2) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(\frac{-(x_i-\mu)^2}{2\sigma^2}\right)$$

log-likelihood function:

$$\ln L(\sigma^2) = -\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln(\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2$$

$$\frac{d}{d(\sigma^2)}\ln L(\sigma^2) = -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2}\sum_{i=1}^{n}(x_i-\mu)^2 = 0$$

$$\sigma^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i-\mu)^2$$

MLE for σ^2 is avg of sq deviation from known mean

Q3) $\bar{x} = 0.042$ mean sd $= 0.01$ $n = 100$

for 99% confidence lev

Z-value $= 2.576$

$$SE = \frac{s}{\sqrt{n}} = \frac{0.01}{\sqrt{100}} = 0.001 \text{ inch}$$

$$E = Z \times SE = 2.576 \times 0.001 = 0.002576$$

lower limit $= \bar{x} - E$ $0.042 - 0.002576$

$= 0.039424$

Upper limit $= \bar{x} + E = 0.042 + 0.002576$

$= 0.044576$

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Section - C

Q1) $L(\theta) = \prod_{i=1}^{5} f(x_i;\theta) = \theta^5 e^{-\theta(x_1+x_2+x_3+x_4+x_5)}$

let $S = x_1+x_2+x_3+x_4+x_5$

then

$$L(\theta) = \theta^5 e^{-\theta S}$$

$$\ln L(\theta) = 5\ln\theta - \theta S$$

$$\frac{d}{d\theta}\ln L(\theta) = \frac{5}{\theta} - S = 0$$

$$\frac{5}{\theta} = S \Rightarrow \theta = 5/S$$

$= 5/0.7+1.5+0.3+0.5+3.4 = 6.4$

$= 5/6.4$

$= 0.78125$

Q2) Likelihood function for binomial distribution

$$L(p) = p^4 (1-p)^{n-4}$$

log likelihood function

$$\ln L(p) = 4\ln(p) + (n-4)\ln(1-p)$$

$$\frac{d}{dp}\ln L(p) = \frac{4}{p} - \frac{n-4}{1-p}$$

$$\frac{d^2}{dp^2}\ln L(p) = -\frac{4}{p^2} - \frac{n-4}{(1-p)^2}$$

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$n_{total} = 50 + 90 = 140$

$k_{total} = 20 + 40 = 60$

likelihood function $L(p) = p^{60}(1-p)^{80}$

$\ln L(p) = 60 \ln p + 80 \ln(1-p)$

$$\frac{d}{dp} \ln L(p) = \frac{60}{p} - \frac{80}{1-p}$$

set derivative to zero for

$$\frac{60}{p} - \frac{80}{1-p} = 0$$

$$60(1-p) = 80p$$

$$60 = 140p$$

$$p = 3/7 = 0.4281$$

$$\frac{d^2}{dp^2} \ln L(p) = -\frac{60}{p^2} - \frac{80}{(1-p)^2}$$

$\because$ both terms are -ve $0 < p < 1$ the log likelihood is concave, $\therefore$ max $p = 3/7$

Q3) Sample mean $= \bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$

$$= \frac{65+71+80+76+78+82+68+72+65+81}{10}$$

$= 73.8$ inches

$s = \sqrt{44.1} = 6.64$

$$SE = \frac{6.64}{\sqrt{10}} = 2.099$$

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$Z_{\alpha/2} = 1.96$

$E = 1.96 \times 2.099 = 4.114$

lower limit $= \bar{x} - E = 73.8 - 4.114 = 69.686$

$\bar{x} + E = 73.8 + 4.114 = 77.914$

Q5) ① $\bar{x} = 160$

$\sigma^2 = 100$

$n = 64$

confidence level $= 95\%$

$s = \sqrt{s^2} = \sqrt{100}$ 10

$z = 1.96$

$$SE = \frac{s}{\sqrt{n}} = \frac{10}{\sqrt{64}} = 1.25$$

$ME = z \times SE = 1.96 \times 1.25 = 2.45$

lower limit $= \bar{x} - ME = 160 - 2.45 = 157.55$

Upper limit $= \bar{x} + ME = 160 + 2.45 = 162.45$

② Desired $ME = 1.4$

confidence level $= 95\%$

$s = 10$

$E = z \times s/\sqrt{n}$

$$n = \left(\frac{z \times s}{E}\right)^2 = \left(\frac{1.96(10)}{1.4}\right)^2 = 196$$

$n_{new} - n_{original}$

$= 196 - 64$

$= 132$

additional observations

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Section - D

6a) (i) $T_1 = x_1 + x_2 - x_3$

$E[T_1] = E[x_1] + E[x_2] - E[x_3]$

$= \mu + \mu - \mu = \mu$

$\therefore$ T_1 is unbiased estimator of μ

$T_2 = 2x_1 + 3x_3 - 4x_2$

$E[T_2] = 2E[x_1] + 3E[x_3] - 4E[x_2] = \cancel{2\mu}$

$= 2\mu + 3\mu - 4\mu$

$= \mu$

$\therefore$ T_2 is a unbiased estimator of μ

(ii) $T_3 = 1/3\,(\cancel{\lambda}\lambda x_1 + x_2 + x_3)$

$E[T_3] = \frac{1}{3}\,[\lambda E(x_1) + E(x_2)] + E(x_3)]$

$= 1/3\,(\lambda\mu + \mu + \mu)$

$= \mu\,\frac{\lambda + \cancel{2}}{3}$

for T_3 to be unbiased

$\frac{\lambda+2}{3} = 1 \Rightarrow \lambda + 2 = 3 \Rightarrow \lambda = 1$

(iii) T_3 with $\lambda = 1$

$T_3 = \frac{1}{3}\,(x_1 + x_2 + x_3)$

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variance, $var(T_3) = \frac{1}{9}\,(var(x_1) + var(x_2) + var(x_3))$

$= \frac{1}{9}\,(3\sigma^2) = \sigma^2/3$

(iv) $var(T_1) = 3\sigma^2$

$var(T_2) = 29\sigma^2$

$var(T_3) = \sigma^2/3$

T_3 has the smallest variance making it the best estimator in terms of variance

(b) likelihood function

$L(\lambda|x) = \prod_{i=1}^{n} \frac{\lambda_i^{x_i} e^{-\lambda}}{x_i!}$

$= \lambda^{\sum x_i} e^{-n\lambda} / \prod_{i=1}^{n} x_i!$

Let $t = \sum x_i$

$L(\lambda|x) \propto \lambda^t e^{-n\lambda}$

Prior distribution

$Gamma(\alpha, \beta) \propto \lambda^{\alpha-1} e^{-\beta\lambda}$

Posterior distribution

$Gamma(\alpha + t, \beta + n)$

Bayesian estimator

$\lambda = \alpha + t / \beta + n$

where $t = \sum x_i$

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Q4) $n = 1000$
$p = 23\%$

$z = 2.576$ for 99% confidence level

$SE = \sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{0.23 \times 0.77}{1000}}$
$= 0.0133$

$ME = z \times SE = 2.576 \times 0.0133 \approx 0.0342$

lower limit $= p - ME = 0.23 - 0.0342$
$= 0.1958$
$= 19.58\%$

upper limit $= p + ME$
$= 0.23 + 0.0342 = 0.2642$
$= 26.42\%$

Q5) $\sigma = 10$
desired $E = 3$
$z = 2.576$ for 99% confidence level

$n = \left(\frac{z \times \sigma}{E}\right)^2$
$= \left(\frac{2.576 \times 10}{3}\right)^2$
$= 73.71 \approx 74$

for $n = 74$
$ME = 2.576 \times \frac{10}{\sqrt{74}}$
$= 2.576 \times 1.1625$
$= 2.993$
(within 3)